习题课4

- 谐振和耦合电路

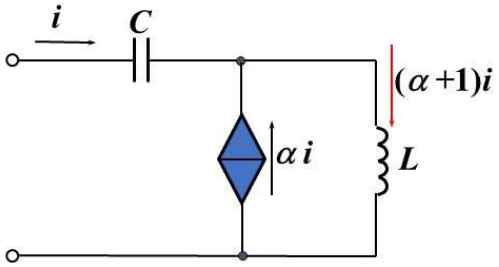
  谐振

  互感

- 第六-七章作业讲评

1. 求谐振频率.

投稿!



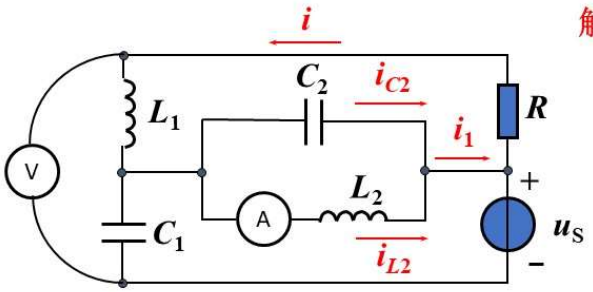
$$\dot{U} = -\mathrm{j}\frac{1}{\omega C}\dot{I} + \mathrm{j}\omega L\dot{I}(1+\alpha)$$

$$Z = \frac{\dot{U}}{\dot{I}} = -\mathrm{j}\frac{1}{\omega C} + \mathrm{j}(1+\alpha)\omega L$$

$$\omega_0 = \frac{1}{\sqrt{(1+\alpha)LC}} \qquad Z(\omega_0) = 0$$

2. 电路如图所示. $u_S(t)=\cos t$ V，$L_1=L_2=1$H，$C_1=C_2=1$F，$R=1\Omega$。

求电压表和电流表的读数(rms).



解：设 $\dot{U}_S=\frac{1}{\sqrt{2}}\angle 0^\circ$ V

$=0.707\angle 0^\circ$ V

$\omega L_2=\frac{1}{\omega C_2}=1\Omega$

(并联谐振)

$\dot{I}_1=0$

$\omega L_1=\frac{1}{\omega C_1}=1\Omega$ (串联谐振),

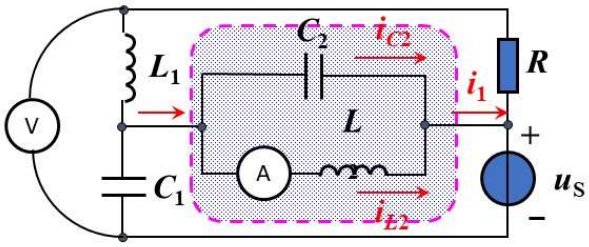
(V)=0

$$\dot{I}=\frac{\dot{U}_S}{R}=\frac{0.707\angle 0^\circ}{1}=0.707\angle 0^\circ \text{ A}$$

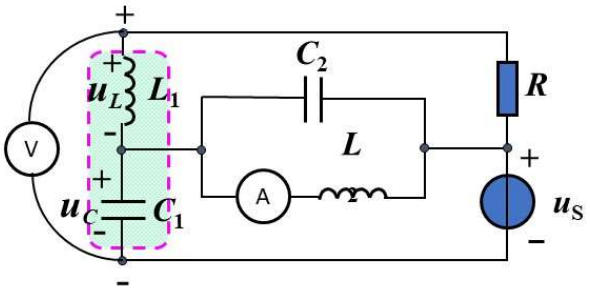
$$\dot{I}_{L2}=\frac{\frac{1}{j\omega C_1}\dot{I}-\dot{U}_S}{j\omega L_2}=\frac{-j0.707-0.707}{j1}=-0.707+j0.707=1.00\angle 135^\circ \text{ A}$$

(A)=1A

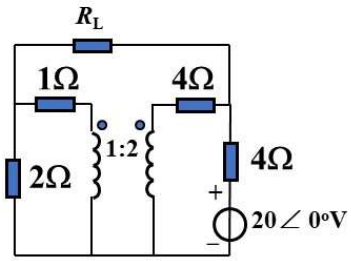
将发生并联谐振的电路看成一个二端网络(或一条广义的支路),则流进或流出端口的电流(或通过该广义支路的电流)为零,但网络内部的各个支路的电流并不一定为零。

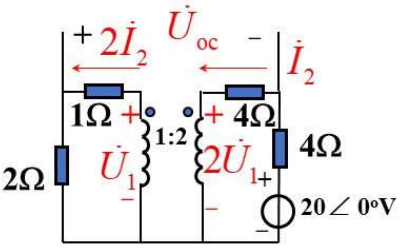


将发生串联谐振的电路看成一个二端网络(或一条广义的支路),则两个端钮之间的电压(或广义支路两端的电压)为零,但网络内部的各个支路上的电压并不一定为零。



3. R_L取值为多大时获得最大功率?最大功率是多少?





(1) 求Uoc

$$3\times 2\dot{I}_2=\dot{U}_1$$

$$8\times\dot{I}_2+2\dot{U}_1=20$$

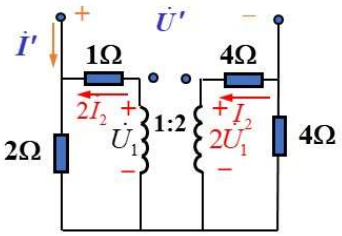
$$\dot{I}_2=1\text{A}$$

$$\dot{U}_{oc}=2\times 2\dot{I}_2+4\dot{I}_2-20$$

$$=-12\text{V}$$

$$\begin{cases}\dot{U}_1=\dfrac{1}{2}\dot{U}_2\\ \dot{I}_1=-2\dot{I}_2\end{cases}\qquad\begin{cases}\dot{U}_2=2\dot{U}_1\\ \dot{I}_1=-2\dot{I}_2\end{cases}$$

（2）加压求流求内阻



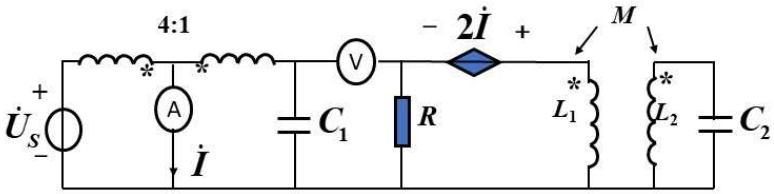
$$\begin{cases} \dot{U}_1 = 2\dot{I}_2 + 2(2\dot{I}_2 + \dot{I}') \\ 4\dot{I}_2 + 2\dot{U}_1 + 4(\dot{I}_2 + \dot{I}') = 0 \\ \dot{U}' = 2(2\dot{I}_2 + \dot{I}') + 4(\dot{I}_2 + \dot{I}') \end{cases} \Rightarrow \dot{U}' = 2.8\dot{I}'$$

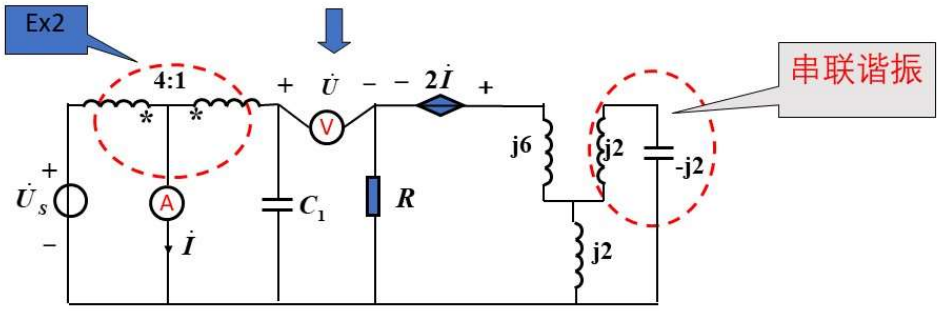
$R = 2.8\Omega$

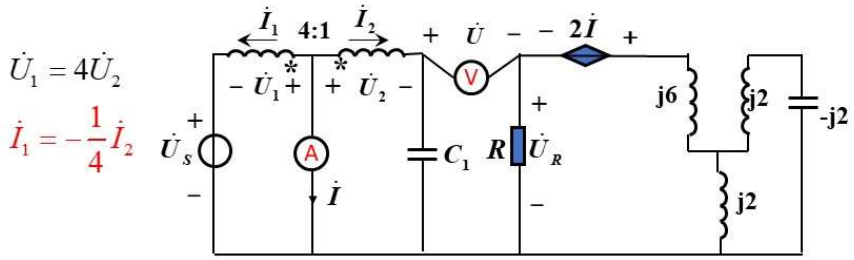
则 R_L＝2.8Ω时获得最大功率.

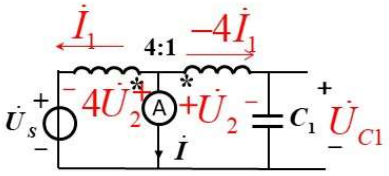
最大功率　P=12²/(4×2.8)=12.86W

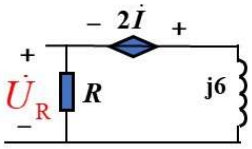
4. $\dot{U}_S = 200\angle 0°, \omega = 2$ rad / s, $C_1 = 0.05$ F, $R = 2\ \Omega$, $L_1 = 4$ H, $L_2 = 2$ H, M = 1H, $C_2 = 0.25$ F　求电压表和电流表的读数。











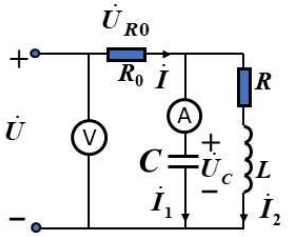
$$\dot{I} = -\,\mathrm{j}3.75\ \mathrm{A}$$

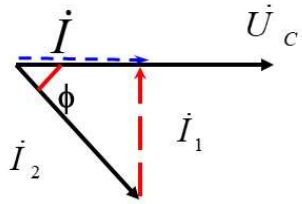
$$\dot{U}_R = \mathrm{j}7.5\,\frac{2}{2+j6} = 2.37\angle 18.43^\circ\ \mathrm{V}$$

$$\dot{U}_{C1} = 50\ \mathrm{V}$$

$$\dot{U} = \dot{U}_{C1} - \dot{U}_R = 47.76\angle -0.9^\circ\ \mathrm{V}$$

5. ω**=1000rad/s时, 电路发生谐振。** R_0**=25Ω,** C**=16μF, 电压表的读数是100V, 电流表的读数是1.2A，求** R **和** L**.**





$$U_C = \frac{1}{\omega C}\times 1.2 = 62.5\times 1.2 = 75\ \mathrm{V}$$

$$U_{R0} = 100 - 75 = 25\ \mathrm{V} \qquad \dot{I} = 1\angle 0^\circ\ \mathrm{A}$$

$$I_2 = \sqrt{1 + 1.2^2} = 1.56\ \mathrm{A}$$

$$\phi = arctg\,\frac{1.2}{1} = -50.19$$

$$R + j\omega L = \frac{\dot{U}_C}{\dot{I}_2} = \frac{75}{1.56\angle -50.19^\circ}$$

$$= 48\angle 50.19^\circ\ \Omega$$

$$R = 30.74\Omega,\ L = 36.88\ \mathrm{mH}$$

6-1、 试求下列正弦电压或电流的振幅、角频率、周期和初相角。

(1) $i_1(t) = 5\cos(2\pi t - 60^\circ)\,A$

(2) $u(t) = 20\sin(\frac{\pi}{6}t - 120^\circ)V$

$\omega T = 2\pi$

解 $u(t) = 20\cos(\frac{\pi}{6}t - 120^\circ - 90^\circ) = 20\cos(\frac{\pi}{6}t + 150^\circ)V$

	振幅	角频率ω(rad/s)	周期T(S)	初相角ψ
(1)	$5\,A$	2π	$1\,S$	-60°
(2)	$20\,V$	$\frac{\pi}{6}$	$12\,S$	150°

6-2、 若已知电流 $i_1(t) = -20\cos(314t + 60^\circ)A$ $i_2(t) = 10\sin(314t + 60^\circ)A$

(1) 写出上述电流的相量，并绘出他们的相量图；
(2) 求出它们的相位差；
(3) 若将i_1表达式中的负号去掉，求该电流的相量？

解

$i_1(t) = 20\cos(314t - 120^\circ)\,A$

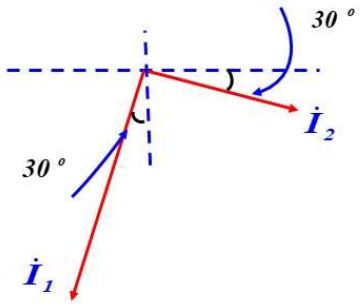
$i_2(t) = 10\cos(314t - 30^\circ)\,A$

(1) $\dot{I}_1 = 10\angle(-120^\circ)$ $\dot{I}_2 = 5\sqrt{2}\angle(-30^\circ)$

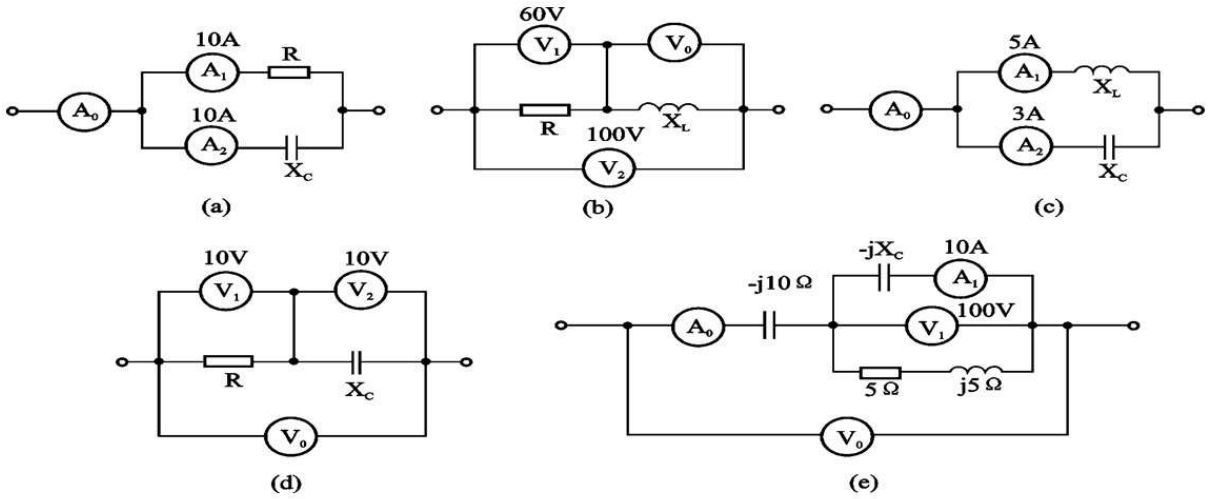
(2) $\varphi = \psi_{i_1} - \psi_{i_2} = -120^\circ - (-30^\circ) = -90^\circ$

(3) $i_1(t) = 20\cos(314t + 60^\circ)\,A$

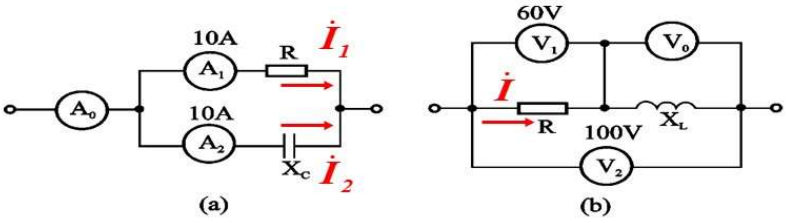
$\dot{I}_1 = 10\angle(60^\circ)$



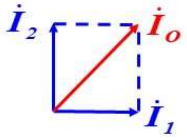
6-3、 在图所示的电路中，除A_0和V_0外，其余电流表和电压表的读数在图上都已标出（均为有效值），试求电流表A_0或电压表V_0的读数。



6-3 、如图所示的电路中，除A_0和V_0外，其余电流表和电压表的读数在图上都已标出（均为有效值），试求电流表A_0或电压表V_0的读数。

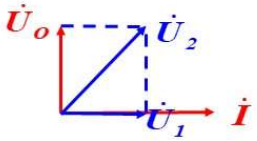


解 （a）

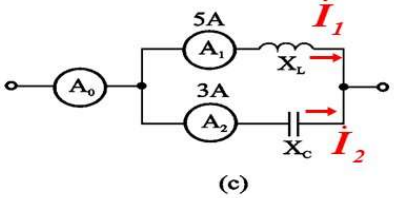


$$I_o = \sqrt{10^2 + 10^2} = 10\sqrt{2}\,A$$

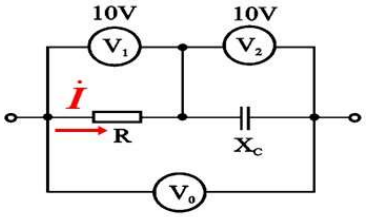
（b）



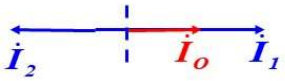
$$U_o = \sqrt{100^2 - 60^2} = 80\,V$$



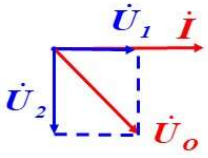
(c)



(d)

解 (c) 

$$I_o = |\dot{I}_1| - |\dot{I}_2| = 2A$$

(d) 

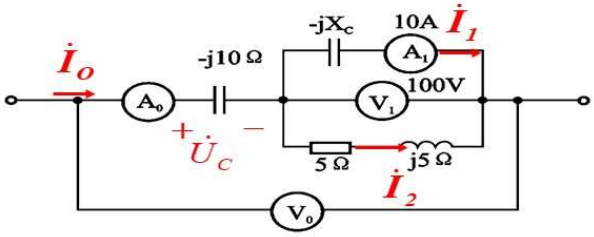
$$U_o = \sqrt{10^2 + 10^2} = 10\sqrt{2}V$$

解 (e) 设 $\dot{U}_1 = 100\angle 0^o$

则 $\dot{I}_1 = 10\angle 90^o$

$Z_2 = 5 + j5(\Omega)$

$\therefore \dot{I}_2 = \dot{U}_1 / Z_2 = 10\sqrt{2}\angle -45^\circ = 10 - j10$

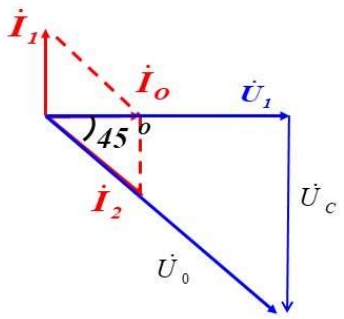


(e)

$\therefore \dot{I}_o = \dot{I}_1 + \dot{I}_2 = j10 + 10 - j10 = 10\ A$

$\dot{U}_C = \dot{I}_o(-j10) = 100\angle(-90^o)$

$\dot{U}_o = \dot{U}_1 + \dot{U}_C = 100\sqrt{2}\angle(-45^o)$



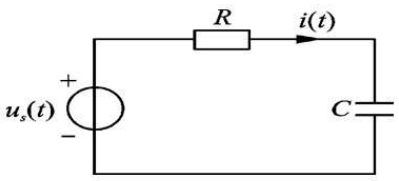
$I_o = 10\ A$

$U_o = 100\sqrt{2}V$

6-4、 电路如图所示，电路处于正弦稳态，其中

$u_s(t) = 10\cos 100t V$，$R = 10\Omega$，$C = 0.001F$，

试求*i*(*t*)。

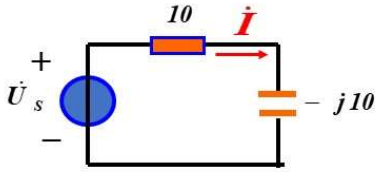


解

$$\dot{U}_S = 5\sqrt{2}\angle 0^0$$

$$\frac{1}{\omega c} = \frac{1}{100 \cdot 0.001} = 10\,\Omega$$

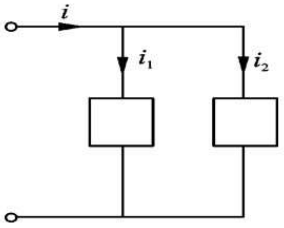
$$\dot{I} = \frac{\dot{U}_S}{R + \frac{1}{j\omega c}} = \frac{5\sqrt{2}}{10 - j10} = \frac{1}{2}\angle 45^o$$



$$\therefore i(t) = \frac{\sqrt{2}}{2}\cos(100t + 45^o)A$$

6-5、 已知图中， $i_1(t) = 10\cos(2\pi t - 36.9^\circ)A$　$i_2(t) = 6\cos(2\pi t + 120^\circ)A$

求*i*(*t*)，并画出相量图。



解

$$\dot{I}_{1m} = 10\angle(-36.9^O) \qquad \dot{I}_{2m} = 6\angle(120^O)$$

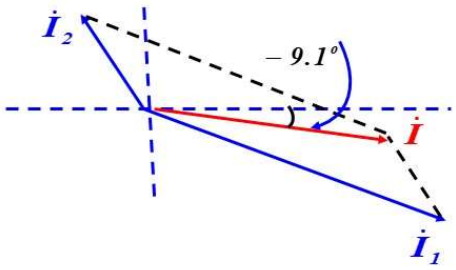
$$\dot{I}_m = \dot{I}_{1m} + \dot{I}_{2m}$$

$$= 10\angle(-36.9^O) + 60\angle(120^O)$$

$$= 8 - j6 + (-3) + j5.2$$

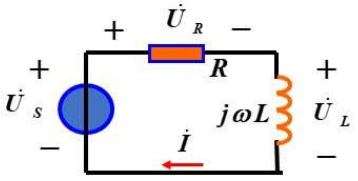
$$= 5 - j0.8$$

$$= 5.1\angle(-9.1^O)$$



$$\therefore i(t) = 5.1\cos(2\pi t - 9.1^o)A$$

6-6、电路由电压源u_s=100 cos(10^3t)V及R和L=0.025H串联组成，电感端电压的有效值为25V。求R值和电流的表达式。

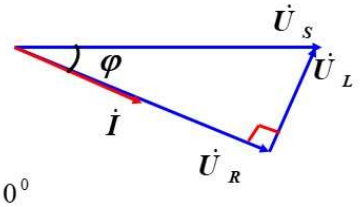


解

$$j\omega L = j10^3 \cdot 0.025 = j25\,\Omega$$

$$\dot{U}_S = 100\angle 0^0$$

$$U_L = 25V \qquad I = \frac{U_L}{\omega L} = \frac{25}{25} = 1A$$



$$U_R = \sqrt{U_S^2 - U_L^2} = \sqrt{(100)^2 - 25^2} = \sqrt{9375} = 96.82V$$

$$\therefore R = \frac{U_R}{I} = \frac{96.82}{1} = 96.82\Omega$$

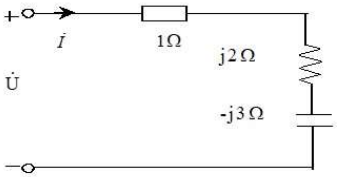
$$\varphi = arctg(\frac{U_L}{U_R}) = 14.48^o$$

$$\dot{U}_S = 100\angle 0^0$$

$$\dot{I} = 1\angle(-14.48^0)$$

$$\therefore i(t) = \sqrt{2}\cos(10^3 t - 14.48^\circ)A$$

7-1、电路图所示的输入阻抗Z和导纳Y。



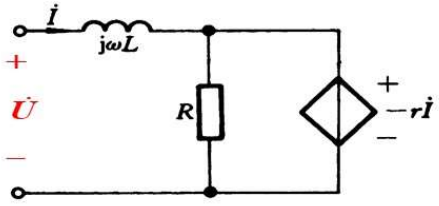
解

$$Z = 1 + j2 - j3$$

$$= 1 - j = \sqrt{2}\angle -45^\circ\,\Omega$$

$$Y = \frac{1}{Z}$$

$$= \frac{1}{\sqrt{2}}\angle 45^\circ = \frac{1}{2} + j\frac{1}{2}(S)$$



$$Z = \frac{\dot{U}}{\dot{I}} = \frac{j\omega L\dot{I} - r\dot{I}}{\dot{I}} = -r + j\omega L$$

$$Y = \frac{1}{Z} = \frac{1}{-r + j\omega L} = \frac{-r - j\omega L}{r^2 + (\omega L)^2}$$

雨课堂 Rain Classroom

7-2、已知下列各负载电压相量和电流相量，试求各负载的等效电阻和等效电抗，并说明负载的性质。

(1) $\dot{U} = (86.6 + j5)V, \dot{I} = (8.66 + j5)A$

(2) $\dot{U} = -100\angle 30^\circ V, \dot{I} = -5100\angle -60^\circ A$

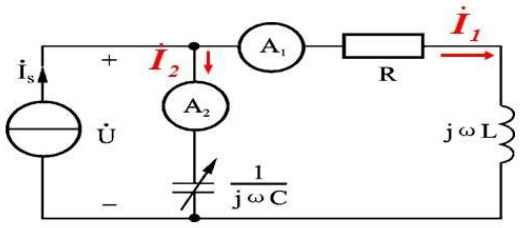
解 **(1)** $Z = \dfrac{\dot{U}}{\dot{I}} = \dfrac{86.6 - j5}{8.66 + j5} = \dfrac{86.7\angle 3.3}{10\angle 30} = 8.67\angle(-26.7)$

$Z = R + jX = 7.74 - j3.89$ 容性

(2) $Z = \dfrac{\dot{U}}{\dot{I}} = \dfrac{-100\angle 30^\circ}{-5100\angle -60^\circ} = \dfrac{1}{51}\angle 90^\circ$

$Z = R + jX = j\dfrac{1}{51}$ 纯感性

7-3、 图示中 $i_S = 14\sqrt{2}\cos(\omega t + \phi)\text{mA}$，**调节电容，使电压** $\dot{U} = U\angle\phi$ **V，电流表A_1的读数为50mA。求电流表A_2的读数。**

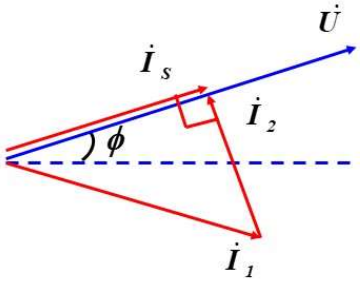


解 $\dot{U} = U\angle\phi$

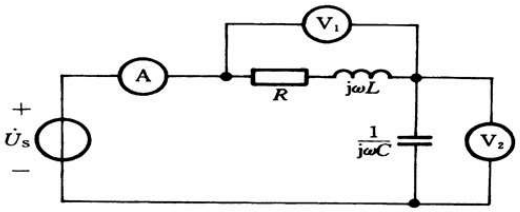
$\dot{I}_S = 14\angle\phi$

显然,U、I_S同相，电路发生谐振。

$I_2 = \sqrt{I_1^2 - I_S^2} = \sqrt{50^2 - 14^2} = 48\,mA$



7-4、图中已知 $u_s = 200\sqrt{2}\cos(314t + \pi/3)V$，电流表A的读数为2A，电压表$V_1$、$V_2$的读数均为200V。求参数$R$、$L$、$C$，并作出该电路的相量图（提示：可先作相量图辅助计算）。



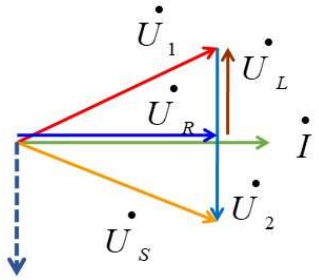
解 作相量图，等边△

$$\dot{I} = 2\angle 90^\circ, \dot{U}_1 = 200\angle 120^\circ$$

$$\dot{U}_R = 100\sqrt{3}\angle 90^\circ = R \times \dot{I} = R \times 2\angle 90^\circ$$

$$\dot{U}_L = 100\angle 180^\circ = j\omega L\dot{I} = j314 \times L \times 2\angle 90^\circ$$

$$\dot{U}_2 = 200\angle 0^\circ = \frac{1}{j\omega C}\dot{I} = \frac{1}{j314\,C} \times 2\angle 90^\circ$$



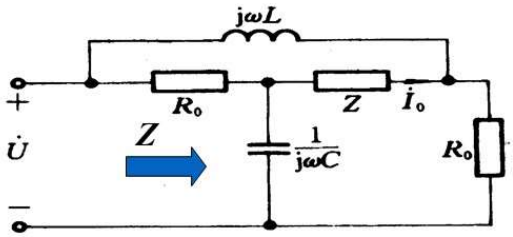
$$\Rightarrow R = 50\sqrt{3}\Omega, L = \frac{25}{157}H, C = \frac{1}{31400}F$$

7-5、已知图示电路中的电压源为正弦量，L=1mH，R_0=1kΩ，Z=（3+j5）Ω。

试求：（1）当$\dot{I}_0 =$时,C值为多少?

（2）当条件（1）满足时，试证明输入阻抗为R_0。

解 （1） **当$\dot{I}_0 = 0$ 时,电路满足电桥平衡。**



$$\frac{j\omega L}{R_O} = \frac{R_O}{1/j\omega C} \Rightarrow R_O^2 = \frac{L}{C}$$

$$\therefore C = \frac{L}{R_O^2} = \frac{10^{-3}}{1000^{\,2}} = 10^3\,pF$$

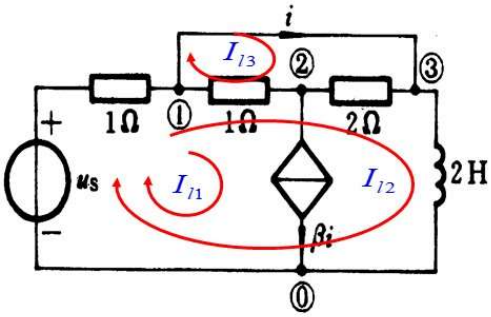
（2）

$$Z = (R_O + j\omega L)//(R_O + \frac{1}{j\omega C}) = \frac{(R_O + j\omega L)(R_O + \frac{1}{j\omega C})}{R_O + j\omega L + R_O + \frac{1}{j\omega C}} = R_O$$

7-6、列出图示电路的回路电流方程和节点电压方程。
已知u_S=14.14cos(2t)V。

解1 **回路法**

$$\begin{cases} \dot{I}_{l1} = \beta\dot{I} \\ 2\dot{I}_{l1} + (4 + j4)\dot{I}_{l2} - 3\dot{I}_{l3} = 10\angle 0^\circ \\ -\dot{I}_{l1} - 3\dot{I}_{l2} + 3\dot{I}_{l3} = 0 \\ \dot{I} = \dot{I}_{l3} \end{cases}$$

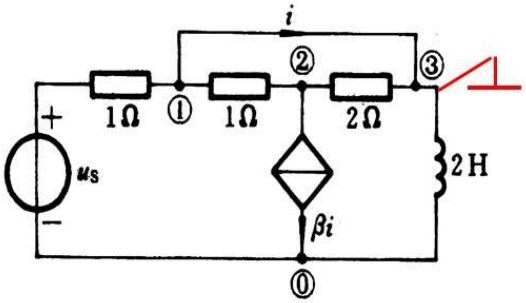


7-6、列出图所示电路的回路电流方程和节点电压方程。
已知u_S=14.14cos(2t)V。

解2 **节点法：**

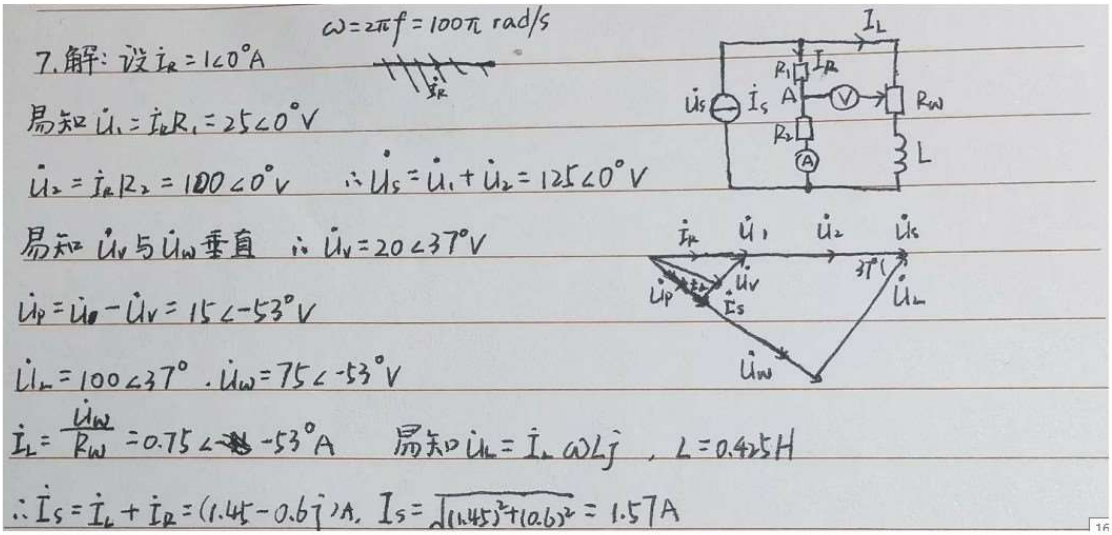
将①③视为参考节点：

$$\begin{cases} (1 + \dfrac{1}{j4})\dot{U}_{n0} = \beta\dot{I} - \dfrac{\dot{U}_S}{1} \\ (1 + \dfrac{1}{2})\dot{U}_{n2} = -\beta\dot{I} \\ \dfrac{\dot{U}_{n2}}{1} + \dfrac{\dot{U}_S + \dot{U}_{n0}}{1} = \dot{I} \end{cases}$$



雨课堂 Rain Classroom

7-7．如图所示正弦稳态电路的工作频率f=50Hz,各表读数为有效值。已知R1=25Ω, R2=100Ω, Rw=100Ω。调节Rw滑动点P使得电压表V读数为最小，且为20V，此时电流表A的读数为1A，试求电流源电流有效值IS及电感L。



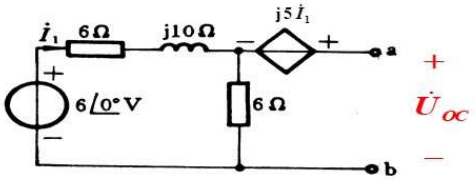
7-7、求图所示一端口电路的戴维宁（或诺顿）等效电路。

解 (1) U_{OC}:

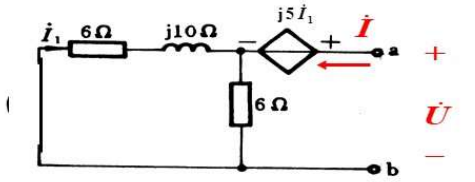
$$6 = (6 + 6 + j10)\dot{I}_1 \qquad \dot{I}_1 = \frac{3}{6 + j5}$$

$$\dot{U}_{OC} = j5\dot{I}_1 + 6\dot{I}_1 = (6 + j5)\frac{3}{6 + j5} = 3\angle 0°$$

$$\therefore \dot{U}_{OC} = 3\angle 0°$$



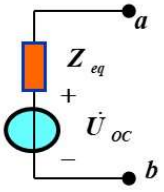
(2) 外加电源法求Zeq:



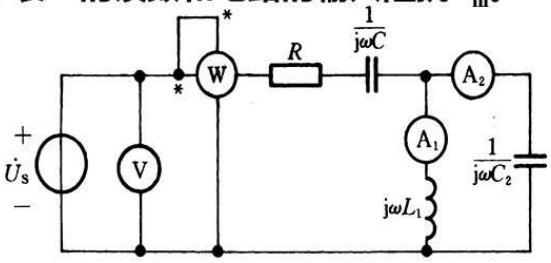
$$\dot{U} = j5\dot{I}_1 + (\dot{I}_1 + \dot{I})6 = (6 + j5)\dot{I}_1 + 6\dot{I}$$

$$(6 + j10)\dot{I}_1 + 6(\dot{I} + \dot{I}_1) = 0 \qquad \dot{I}_1 = \frac{-3\dot{I}}{6 + j5}$$

$$\therefore \dot{U} = (6 + j5)\frac{-3\dot{I}}{(6 + j5)} + 6\dot{I} = 3\dot{I} \qquad \therefore Z_{eq} = \frac{\dot{U}}{\dot{I}} = 3\,\Omega$$



7-8、图所示电路中已知：$1/\omega c = 1.5\omega L_1$，$R=1\Omega$，$\omega=10^4 rad/s$，电压表的读数为10V，电流表$A_1$的读数为30A。求图中电流表$A_2$、功率表W的读数和电路的输入阻抗$Z_{in}$。



解 设：$\dot{I}_1 = 30 \angle 0^\circ$

$$\dot{I}_2 = \frac{j\omega L_1 \dot{I}_1}{1/(j\omega C_2)} = \frac{j\omega L_1 \dot{I}_1}{-j1.5\omega L_1}$$

$$= \frac{-\dot{I}_1}{1.5} = -20 \angle 0^\circ \qquad \boxed{I_2 = 20\ A}$$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = 30 - 20 = 10 \angle 0^\circ$$

$$\boxed{P = RI^2 = 1 \times 10^2 = 100\ W}$$

$$\cos\varphi = \frac{P}{UI} = \frac{100}{10 \times 10} = 1 \qquad \varphi = 0^\circ \quad \text{即 } \dot{U}、\dot{I} \text{同相（串联谐振）}$$

$$\Rightarrow \quad \boxed{\therefore Z_{in} = \frac{\dot{U}}{\dot{I}} = \frac{10}{10} = 1\Omega}$$

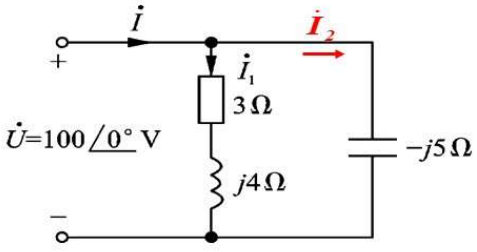
7-9、一端口网络的相量模型如图所示，求电源发出的复功率。

解

$$\dot{I}_1 = \frac{\dot{U}}{Z_1} = \frac{100}{3+j4} = 12 - j6 = 20 \angle (-53.1^\circ)$$

$$\dot{I}_2 = \frac{\dot{U}}{-j5} = 20j = 20 \angle 90^\circ$$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = 12 - j16 + j20 = 12.65 \angle 18.43^\circ$$



方法1

$$P = UI\cos(0 - 18.43) = 100 \cdot 12.65 \cdot \cos(-18.43) = 1200\ W$$

$$Q = UI\sin(0 - 18.43) = 100 \cdot 12.65 \cdot \sin(-18.43) = -400\ Var$$

$$\hat{S} = P + jQ = 1200 - j400\ (V \cdot A)$$

方法2

$$\hat{S} = \dot{U}\dot{I}^* = 100 \cdot 12.65 \angle (-18.43) = P + jQ = 1200 - j400\ (V \cdot A)$$

7-10、已知图所示电路中，I_1=10A，I_2=20A，其功率因数分别为

$\lambda_{1=}\cos\varphi_1 = 0.8(\varphi_1 < 0)$　　$\lambda_2 = \cos\varphi_2 = 0.5(\varphi_2 > 0)$

端电压U=100V，ω=1000rad/s。（1）求图中电流表、功率表的读数和电路的功率因数；（2）若电源的额定电流为30A，那么还能并联多大的电阻？求并联该电阻后功率表的读数和电路的功率因数；（3）如使原电路的功率因数提高到λ=0.9，需要并联多大电容？

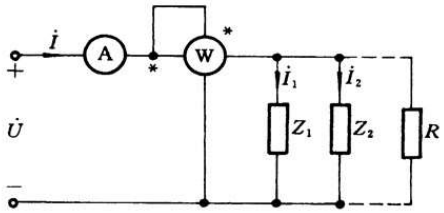
解

(1)

$P_1 = UI_1\cos\varphi_1 = 100\times10\times0.8 = 800W$

$P_2 = UI_2\cos\varphi_2 = 100\times20\times0.5 = 1000\ W$

W　$P = P_1 + P_1 = 1800\ W$



$\dot{U} = 100\angle0^\circ V$

$\varphi_1 = \cos^{-1}0.8 = -36.87^\circ$　　$\dot{I}_1 = 10\angle36.87^\circ$

$\varphi_2 = \cos^{-1}0.5 = 60^\circ$　　$\dot{I}_2 = 20\angle(-60^\circ)$

$\dot{I} = \dot{I}_1 + \dot{I}_2 = 10\angle36.87^\circ + 20\angle(-60^\circ) = 21.26\angle-32.16^\circ$

$\cos\varphi = \cos[0-(-32.16^\circ)] = 0.846$

（2）若电源的额定电流为30A，那么还能并联多大的电阻？求并联该电阻后功率表的读数和电路的功率因数；

解　(2) **并联电阻后:**

$\dot{I}' = \dot{I}_1 + \dot{I}_2 + \dot{I}_R = 18 - j11.32 + \frac{100}{R}$

$\because (18+\frac{100}{R})^2 + 11.32^2 = 30^2 \Rightarrow R = 10.22\ \Omega$

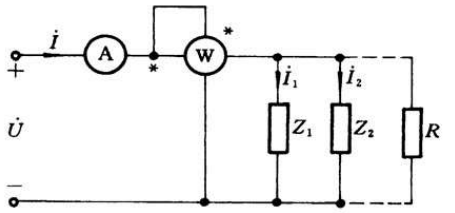
$\therefore\ \dot{I} = (18+\frac{100}{R}) - j11.32$

$= 27.78 - j11.32 = 30\angle(-22.17^\circ)$

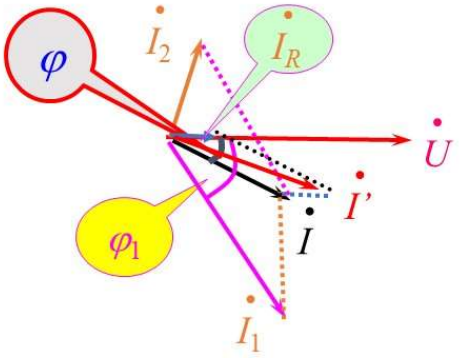
$\cos\varphi = \cos(22.17^\circ) = 0.926$

$P_R = \frac{U^2}{R} = \frac{100^2}{10.22} = 978.5W$

$\therefore P = P_1 + P_2 + P_R = 2778.5W$

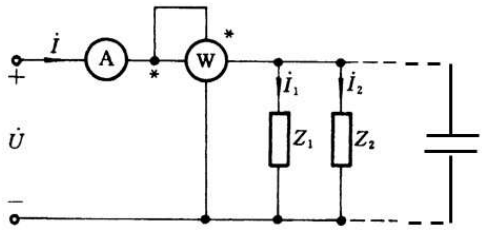


$\dot{U} = 100\angle0^\circ V$



(3) 如使原电路的功率因数提高到λ=0.9，需要并联多大电容?

解 **并联电容后:**



$$\varphi' = \cos^{-1} 0.9 = 25.84^\circ$$

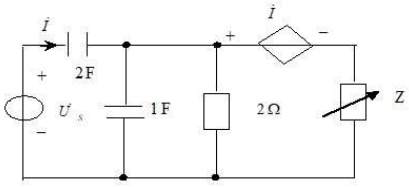
$$\because \quad C = \frac{P}{\omega U^2}(tg\,\varphi - tg\,\varphi')$$

$$= \frac{1800}{1000 \times 100^2}(tg\,32.17 - tg\,25.84)$$

$$= \frac{1800}{10^7}(0.628 - 0.484) = 25.8\,\mu F$$

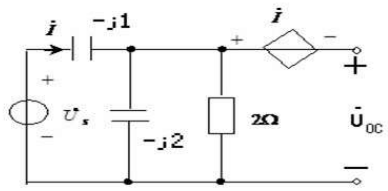
7-11、如图示电路，已知 $u_S = 2\sqrt{2}\cos(0.5t + 120^\circ)V$ **Z可变。**

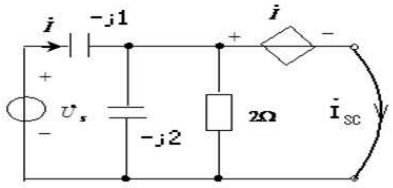
求Z为何值时，可获最大功率P_{max}，并求P_{max}的值。

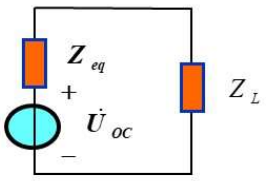


解: (1) $\dot{U}_{OC}$ $\quad \because (-j2\Omega \,\|\, 2\Omega = 1 - j)$

$$\dot{I} = \frac{\dot{U}_s}{-j+1-j} = \frac{2\angle 120^\circ}{1-j2}, \dot{U}_{OC} = (1-j1)\times\dot{I} - \dot{I} = \frac{2}{\sqrt{5}}\angle 93.4^\circ$$







(2) $\dot{I}_{SC}$: $\quad -j1\times\dot{I} + \dot{I} = \dot{U}_S \Rightarrow \dot{I} = \sqrt{2}\angle 165^\circ \Rightarrow \dot{I}_{SC} = \dot{I} - \dot{I}/(-j2) - \dot{I}/2 = 1\angle 120^\circ$

$$\Rightarrow Z_{eq} = \dot{U}_{SC}/\dot{I}_{SC} = \frac{2}{\sqrt{5}}\angle -26.6^\circ\Omega$$

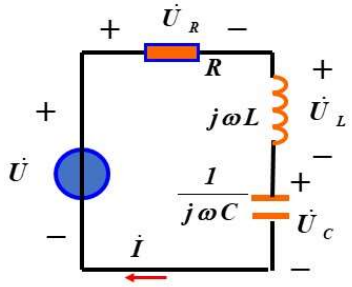
(3) $Z_L = Z_{eq}^*, \Rightarrow P_{\max} = \frac{U_{OC}^2}{4R_{eq}} = 0.25W$

7-12、当ω=5 000rad/s时，RLC串联电路发生谐振，已知R=5Ω，L=400mH，端电压U=1V。求电容C的值及电路中的电流和各元件电压的瞬时表达式。

解 **发生谐振时：** $\because \omega = \dfrac{1}{\sqrt{LC}}$

$$\therefore C = \frac{1}{\omega^2 L} = \frac{1}{5000^2 \times 400 \cdot 10^{-3}} = 0.5\mu F$$

若 $u(t) = u_R(t) = \sqrt{2}\cos 5000\ t$



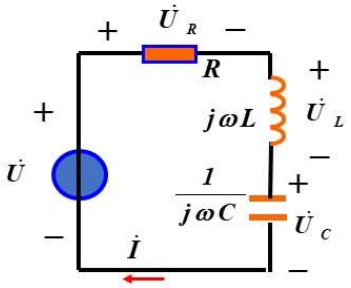
$$\dot{I} = \frac{\dot{U}}{R} = 0.2\angle 0^\circ \qquad \therefore\ i(t) = 0.2\sqrt{2}\cos 5000\ t$$

$$U_L = U_C = \omega L I = 5000 \times 0.4 \times 0.2 = 400\,V$$

$$u_L(t) = 400\sqrt{2}\cos(5000\ t + 90^\circ) \qquad u_C(t) = 400\sqrt{2}\cos(5000\ t - 90^\circ)$$

7-13．RLC串联电路的端电压u=10 cos(2 500t+10°)V，当C=8μF时，电路中吸收的功率为最大，P_{max}=100W。

（1）求电感L和Q值；（2）作出电路的相量图。

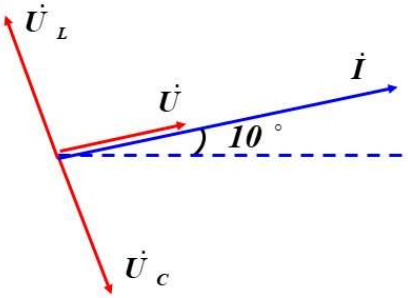


解 **当有P=Pmax时，电路发生串联谐振**

$$\frac{1}{\omega C} = \omega L$$

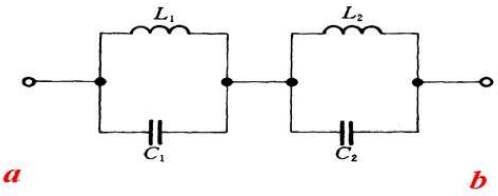
$$\therefore L = \frac{1}{\omega^2 C} = \frac{1}{2500^2 \times 8 \cdot 10^{-6}} = \frac{1}{50}H$$

$$Q = \frac{\omega L}{R} = \frac{2500 \cdot \frac{1}{50}}{1} = 50$$



雨课堂 Rain Classroom

7-14、求图6-11电路的谐振频率。



解 $$Z_{ab} = j\omega L_1 // \frac{1}{j\omega C_1} + j\omega L_2 // \frac{1}{j\omega C_2}$$

$$= \frac{j\omega L_1}{1-\omega^2 L_1 C_1} + \frac{j\omega L_2}{1-\omega^2 L_2 C_2}$$

谐振时：　Z_{ab}的虚部为0，即Zab=0

$$\frac{\omega L_1}{1-\omega^2 L_1 C_1} + \frac{\omega L_2}{1-\omega^2 L_2 C_2} = 0$$

$$\omega = \sqrt{\frac{L_1 + L_2}{L_1 L_2 (C_1 + C_2)}}$$

当Z_{ab}=∞时，发生并联谐振

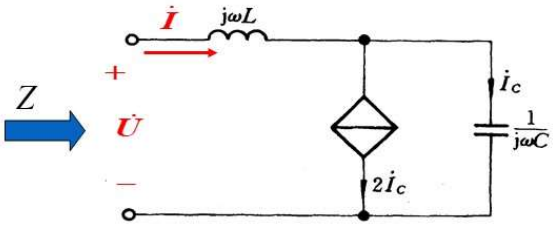
$$\omega L_1 - \frac{1}{\omega C_1} = 0$$

$$\omega_o = \frac{1}{\sqrt{L_1 C_1}}$$

$$\omega L_2 - \frac{1}{\omega C_2} = 0$$

$$\omega_o = \frac{1}{\sqrt{L_2 C_2}}$$

7-14、求图示电路的谐振频率。



解：(外加电源)

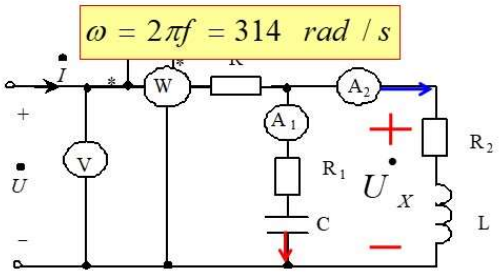
$$\dot{I} = 2\dot{I}_C + \dot{I}_C = 3\dot{I}_C$$

$$\dot{U} = j\omega L\dot{I} + \frac{1}{j\omega C}\dot{I}_C = j\omega L\dot{I} + \frac{1}{j\omega C}\frac{\dot{I}}{3}$$

$$\therefore Z = \frac{\dot{U}}{\dot{I}} = j(\omega L - \frac{1}{3\omega C})$$

谐振时，有：　$\omega L - \frac{1}{3\omega C} = 0$ ⇒ $\omega_o = \frac{1}{\sqrt{3LC}}$

7-15、电路如图所示。已知电流表A_1，A_2的读数为10A，电压表的读数为220V(均为有效值)，功率表的读数为2200W，R=12Ω，电源频率f=50Hz，且已知 $\dot{U}$, $\dot{I}$ 同相。试求R_1，R_2，L和C。



解 设 $\dot{U} = 220\angle 0^\circ$， $\dot{I} = I\angle 0^\circ, \varphi = 0^\circ$

$2200 = P = UI\cos\varphi \quad \Rightarrow I = 10$

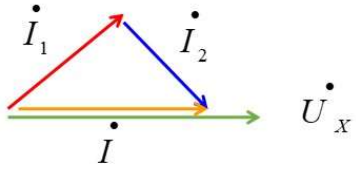
作相量图，等边△

$\dot{I}_1 = 10\angle 60^\circ$， $\dot{I}_2 = 10\angle -60^\circ$

$\dot{U}_X = \dot{U} - R\dot{I} = 100\angle 0^\circ$

$R_1 + \dfrac{1}{j\omega C} = \dfrac{\dot{U}_X}{\dot{I}_1} = 10\angle -60^\circ$

$R_2 + j\omega L = \dfrac{\dot{U}_X}{\dot{I}_2} = 10\angle 60^\circ$



$\Rightarrow R_1 = R_2 = 5\Omega, L = \dfrac{5\sqrt{3}}{314}H, C = \dfrac{1}{1570\sqrt{3}}F$

得 分	
评卷人	

二、如图 2 所示的直流电路，用节点电压法求电压 U、电流 I 及两个受控源的功率。(12 分)

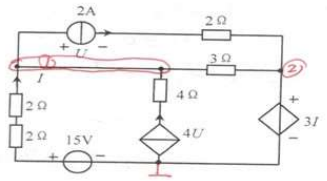


图 2

解：如图标注，有

$(\frac{1}{3}+\frac{1}{4})U_{n1} - \frac{1}{3}U_{n2} = \frac{15}{4} + 4U - 2$

$U_{n2} = 3I$

$U = U_{n1} - U_{n2} - (2\times 2)$

$I = -\dfrac{U_{n1} - 15}{4}$

$P_{4U} = (U_{n1} + 4\times 4U)\times 4U$ (认为发)

$P_{3I} = (2 + \dfrac{U_{n1} - 3I}{3})\times 3I$ (认为吸)

解得：$U =$ $I =$

$P_{3I} =$ $P_{4U} =$

得分	
评卷人	

六、直流电路如图 6 所示，N_S 是线性含源电阻网络。请根据图 6(a)和(b)的结果，求（1）图 6(c)中 I_1 的值；（2）图 6(d)中 R_L 为何值时其可获得最大功率？并求此最大功率。(12 分)

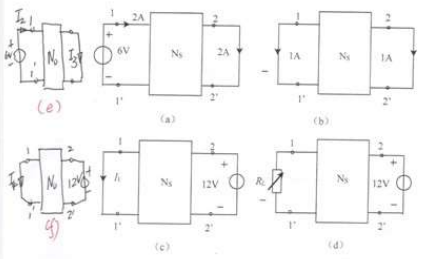


图 6

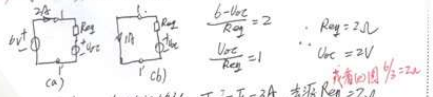
解：(1) N_0 是 N_S 去源后的无源线性电阻网络

∵ (a) = (b) + (e)　∴ $I_2 = 2 + 1 = 3A$，$I_3 = 2 - 1 = 1A$

(e) 反向 互易+齐性　∴ $I_4 = \frac{12}{6} \times I_3 = 2A$

∵ (c) = (b) + (f)　∴ $I_1 = 1 + I_4 = 3A$

(2) 对(a)、(b)方例 1-1' 等效



$\frac{6 - U_{oc}}{R_{eq}} = 2$　$\frac{U_{oc}}{R_{eq}} = 1$　∴ $R_{eq} = 2\Omega$，$U_{oc} = 2V$

对(d)方例 1-1' 诺顿等效　$I_{sc} = I_1 = 3A$，去源 $R_{eq} = 2\Omega$　并联电阻 $\frac{6}{3} = 2\Omega$

即 R_L



∴ $R_L = 2\Omega$ 时

$P_{max} = (\frac{1}{2} I_{sc})^2 R_L = 4.5W$